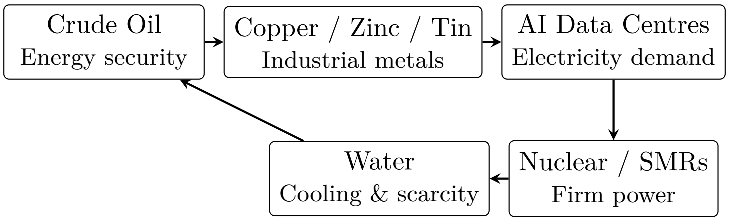
From Dark Gold to Digital Gold

The Commodities Behind the AI–Energy Transition

A research note on crude oil, industrial metals, AI data centres, nuclear power, cooling infrastructure and water



Core thesis. The original idea behind this note is that the next major commodity opportunity may emerge from the physical infrastructure required by the digital economy. Crude oil remains strategically important, but the rapid expansion of AI can create additional demand for electricity, grids, industrial metals, reliable generation, cooling infrastructure and water.
The objective of this report is therefore to examine the chain

$$\text{AI} \rightarrow \text{Electricity} \rightarrow \text{Infrastructure} \rightarrow \text{Resources}$$

and identify where economic scarcity could migrate next.

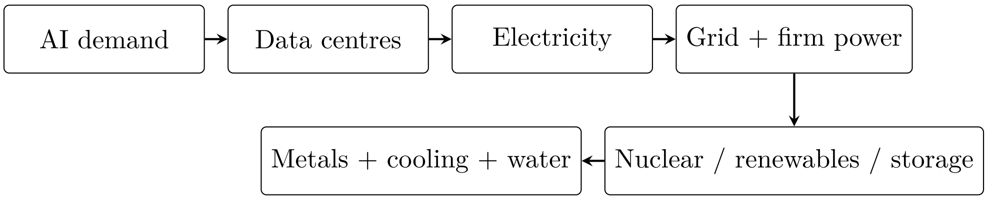
September 2026

1. Executive Summary

The original research note began with uncertainty in the crude-oil market following U.S. sanctions on major Russian oil companies, before moving towards industrial metals, AI data centres, small modular reactors (SMRs), and finally water. The sequence is retained here because it provides the central investment question: *where does economic scarcity migrate when demand shifts towards a new physical infrastructure system?*

The strongest interpretation of the thesis is not that every commodity price increase is caused by AI. Instead, AI can create a new layer of physical demand. Data centres require computing hardware, electricity, transmission capacity, substations, transformers, cooling systems and reliable power.

This creates a broader resource chain:



India is particularly relevant because the country is simultaneously expanding its digital infrastructure, electricity system and nuclear programme. The Government of India has stated a target of 100 GW of nuclear capacity by 2047, while also discussing SMRs for applications including energy-intensive industries and repurposing retiring fossil-fuel sites.

The investment question is therefore broader than a single commodity:

> **The next commodity opportunity may emerge wherever AI-driven physical demand meets a constrained resource.**

2. From Crude Oil to a Broader Commodity Thesis

The source draft begins with uncertainty surrounding Russian crude flows after sanctions on Rosneft and Lukoil. It also discusses pressure on Asian crude demand, lower refining margins in China, and uncertainty surrounding tankers carrying Russian crude towards India. These observations are the starting point of the original thesis, not its final conclusion.

The more useful question is:

Where does economic scarcity migrate when one strategic resource becomes constrained?

Crude oil remains central to transportation, petrochemicals and industrial activity. However, the rise of artificial intelligence introduces another form of resource intensity.

A high-density data centre requires:

- computing hardware and semiconductors;
- high-voltage grid connections;
- substations and transformers;
- reliable electricity generation;
- cooling infrastructure;
- land, fibre and network connectivity.

Consequently, the commodity map can expand from

$$\text{Oil} \rightarrow \text{Transportation}$$

towards

$$\text{Electricity} \rightarrow \text{Compute} \rightarrow \text{Infrastructure.}$$

This does not mean that oil becomes irrelevant. It means that investors should track several potential bottlenecks rather than assuming that the next commodity cycle must be dominated by crude oil.

3. The Original Observation: Copper, Zinc, Tin and Silver

The original draft observed that from April 2025 there was notable price action in copper, zinc, tin and silver. It associated silver and copper with semiconductor and Chinese infrastructure demand, respectively, and proposed that zinc and tin could benefit from the growing energy requirements associated with AI data centres.

This observation is interesting, but the causal relationship should be treated as a **research hypothesis** rather than an established fact.

A metal can rise because of several factors simultaneously:

- changes in physical inventories;
- Chinese industrial demand;
- semiconductor and electronics demand;
- mine supply disruptions;
- smelter economics;
- investment flows;
- expectations about future infrastructure spending.

The correct research question is therefore:

> **Which materials become economically scarce if data-centre capacity grows faster than power, grid and cooling infrastructure?**

This is a testable question and provides a better foundation for the commodity thesis.

4. Copper, Zinc, Tin and Silver: Different Exposures

The four metals should not be treated as one homogeneous "AI metals" basket.

Metal	**Relevant exposure**	**Variables to test**
Copper	Power cables, transformers, motors, electrical infrastructure and data-centre buildout	Mine supply, inventories, grid investment, construction demand, China and recycling
Zinc	Galvanisation and corrosion protection across infrastructure and industrial equipment	Steel demand, construction, China, mine supply, treatment charges and inventories
Tin	Soldering/electronics and other industrial applications; nuclear applications require separate analysis	Electronics demand, solder intensity, supply concentration, inventories and substitution
Silver	Electronics, electrical applications, solar and investment demand	Industrial demand, photovoltaic growth, investment flows, mine supply and gold/silver ratio

Table 1: Research framework for the four metals.

Copper is arguably the cleanest first-order infrastructure expression of this thesis because electricity transmission and distribution are copper-intensive.

Zinc and tin are more nuanced. Their prices should not be attributed to AI without a proper end-use demand analysis.

The next step should therefore be quantitative: construct an indexed chart with **April 2025 = 100** and compare copper, zinc, tin, silver and crude oil.

The chart should then be studied alongside:

- inventory changes;
- Chinese industrial activity;
- semiconductor and electronics indicators;
- global data-centre capacity announcements;
- electricity-demand forecasts.

This would turn the original visual observation into a falsifiable research test.

5. The AI–Electricity Bottleneck

The most defensible part of the original thesis is the relationship between AI and electricity demand.

The source draft includes a discussion of the rapid expansion of generative AI and the resulting requirement for high-density data centres and significantly more electricity to power them. It also reproduces a Goldman Sachs Research statement about projected growth in data-centre power demand.

The underlying mechanism is straightforward:

$$\boxed{\text{More compute}} \Rightarrow \boxed{\text{More electricity}} \Rightarrow \boxed{\text{More generation + grid investment}}$$

The infrastructure requirement extends beyond electricity generation. Data-centre growth also creates demand for:

1. transformers;
2. transmission and distribution equipment;
3. electrical conductors;
4. cooling equipment;
5. backup power;
6. construction materials;
7. network and fibre infrastructure.

This is the point at which the AI story becomes a commodity story.

6. Nuclear Power and the SMR Optionality

The original draft then asks why zinc and tin could matter in a nuclear-powered future. It notes that zinc can be used in corrosion-control applications in coolant systems and that tin has applications in fuel-rod cladding materials.

These applications are worth investigating, but they should not by themselves be used to explain commodity-price movements. The relevant question is whether future reactor deployment would create material demand large enough to affect global supply-demand balances.

The more important macro development is the renewed interest in nuclear power as a source of reliable electricity.

The source draft argues that SMRs could become relevant to the energy requirements of AI data centres. The stronger formulation is that **some developers, utilities and policymakers are exploring nuclear and SMR power as potential sources of reliable electricity for growing loads**. It would be too strong to claim that all AI data centres are moving towards SMRs.

6.1 India's Nuclear Opportunity

India provides an important case study.

The government's nuclear roadmap includes a target of:

100 GW of nuclear capacity by 2047

The present nuclear capacity is around 8.78 GW according to the government material used for this research. The roadmap also includes indigenous SMR development.

The announced Indian designs include:

- **BSMR-200:** 220 MWe;
- **SMR-55:** 55 MWe;
- **HTGCR:** up to 5 MWth for heat applications including hydrogen production.

The government has also discussed a target of developing and operationalising multiple indigenous SMRs.

The key investment question therefore becomes:

> **Which nuclear technologies can deliver reliable power at a cost and construction speed compatible with rapidly growing electricity demand?**

That is a more useful question than simply asking whether nuclear will replace fossil fuels.

6.2 The Execution-Risk Problem

The source draft correctly highlights the difficulty of moving nuclear technology from design or prototype stages into repeatable commercial deployment.

The lessons from large nuclear projects, including the Vogtle expansion in the United States, illustrate the importance of:

- regulatory approvals;
- construction productivity;
- supply-chain reliability;
- financing costs;
- manufacturing capacity;
- project standardisation;
- electricity offtake agreements.

SMRs are often promoted on the basis of standardisation and factory manufacturing. However, those benefits are not automatically realised at commercial scale.

Therefore:

> **SMR is an infrastructure-execution thesis, not merely a reactor-design thesis.**

The original draft also refers to the U.S. administration's efforts to revive nuclear development and to a reported strategic partnership involving Westinghouse and Cameco. That development is relevant as evidence of renewed nuclear interest, but the exact project structure, investment amount and implementation timeline should be independently verified before being used as an investment assumption.

7. Water: The Most Interesting Second-Order Thesis

Water is the most provocative part of the original report.

The source draft argues that AI data centres and nuclear facilities can require substantial quantities of water for cooling, and therefore proposes water as a potential long-term commodity opportunity.

The underlying intuition is valid: **cooling is a physical constraint**. However, water use is highly dependent on technology.

Data-centre water consumption can vary with:

- cooling architecture;
- climate;
- operating conditions;
- heat-rejection technology;
- water-recycling systems.

The same principle applies to nuclear facilities.

It is therefore too broad to state that all data centres require large quantities of potable freshwater or that saline water is universally unusable. The investment thesis should instead focus on **water stress and cooling infrastructure**.

The stronger chain is:

$$\text{AI growth} \rightarrow \text{Data-centre construction} \rightarrow \text{Cooling demand} \rightarrow \text{Water infrastructure} \rightarrow \text{Potential scarcity premium}$$

7.1 Can Water Actually Be Traded?

The original draft ends by asking whether water can be traded through CME and notes that it could not identify a sufficiently clear source explaining how water could be traded as a commodity.

There is an exchange-traded water-price derivative, but the market is highly specific.

CME lists futures based on the:

Nasdaq Veles California Water Index (NQH2O)

The index is connected to water-rights transactions in major California markets and is not a generic global water-price contract.

Therefore, the research conclusion should be:

> **There is an exchange-traded water-price derivative, but there is not a generic global "water futures" market equivalent to crude oil futures.**

This distinction is important.

7.2 The More Interesting Water Trade

Rather than immediately treating a water derivative as the investment opportunity, the broader infrastructure ecosystem may be more useful to study.

Potential areas include:

- water treatment;
- industrial water recycling;
- desalination;
- cooling technology;
- closed-loop cooling;
- water infrastructure;
- municipal supply systems;
- legally investable water rights.

This creates a second-order investment thesis:

$$\boxed{\text{AI} \to \text{Power} \to \text{Cooling} \to \text{Water Infrastructure}}$$

The opportunity is therefore not necessarily “buy water.” It may be to identify the companies and infrastructure that become valuable when water becomes a binding constraint on digital infrastructure.

8. India: Why the Thesis Matters Locally

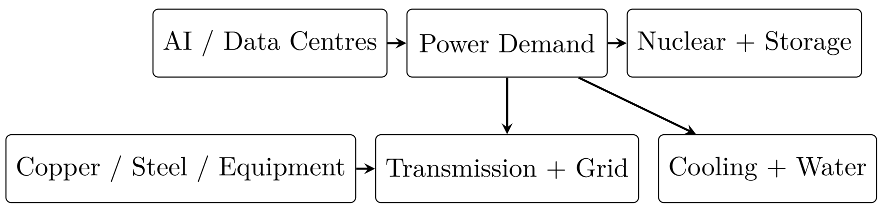
India should not appear in this report merely as a political footnote.

It is a useful case study because the country is simultaneously trying to expand:

- digital infrastructure;
- data-centre capacity;
- electricity generation;
- transmission infrastructure;
- nuclear capacity.

The government’s nuclear roadmap targets 100 GW by 2047, compared with a present capacity of roughly 8.78 GW.

This creates a potential domestic research basket:



The opportunity is therefore broader than a single policy bet. It is an attempt to identify the physical infrastructure required for the digital economy.

9. How to Turn the Thesis Into an Investable Research Model

The original narrative can eventually be converted into a quantitative framework.

A useful five-layer scorecard is:

Layer	Variables	Potential signal
AI demand	Data-centre capacity, GPU deployment, cloud capex	Accelerating capacity additions
Electricity	Peak demand, generation additions, grid capex	Persistent demand growth with constrained supply
Metals	Prices, inventories, treatment charges, mine supply	Tightening inventories and supply deficit
Nuclear	Reactor approvals, construction starts, capex	Rising project pipeline with credible financing
Water	Water stress, cooling intensity, recycling investment	Increasing scarcity with infrastructure spending

Table 2: Suggested quantitative monitoring framework.

A composite signal could be written as:

$$S_t = w_1 Z(\Delta AI_t) + w_2 Z(\Delta Power_t) + w_3 Z(MetalTightness_t) + w_4 Z(NuclearPipeline_t) + w_5 Z(WaterStress_t),$$

where $Z(\cdot)$ represents a standardised variable and:

$$\sum_{i=1}^{5} w_i = 1.$$

The model could then be back-tested against:

- commodity returns;
- mining-equity returns;
- power-equipment companies;
- utilities;
- water-infrastructure companies.

This is the point at which the report can move from a narrative macro thesis towards a quantitative commodity signal.

10. Key Risks to the Thesis

A serious research report should argue against its own thesis.

1. **AI efficiency risk:** Better chips, model compression and inference efficiency could reduce electricity consumption per unit of compute.
2. **Grid bottleneck risk:** Data-centre demand could be constrained by transmission capacity rather than generation capacity.
3. **Nuclear execution risk:** Regulation, financing and construction delays could prevent projected capacity from arriving on schedule.
4. **Metal substitution risk:** Engineering changes could reduce material intensity or substitute one material for another.
5. **Water-technology risk:** Closed-loop, immersion and other cooling technologies could reduce freshwater intensity.
6. **Valuation risk:** A correct structural thesis can still produce poor investment returns if expectations are already fully priced.
7. **Commodity-cycle risk:** China, global manufacturing cycles and inventories can dominate short-term metal prices even when long-term AI demand is rising.

The most important principle is:

> **A structural demand story is not the same thing as a profitable trade.**

11. Conclusion

The original "Dark Gold → Digital Gold" idea contains a useful macro intuition: the world's most important resources may increasingly be determined by the infrastructure required to operate the digital economy.

Crude oil remains strategically important. However, AI introduces a new physical resource stack:

$$\boxed{\text{Compute} \rightarrow \text{Electricity} \rightarrow \text{Generation} \rightarrow \text{Grid} \rightarrow \text{Metals} \rightarrow \text{Cooling} \rightarrow \text{Water}}$$

India provides an especially useful case study because its policy framework is simultaneously expanding digital infrastructure and targeting a substantial increase in nuclear capacity.

The strongest conclusion is therefore not:

> "Water is the next oil."

Instead:

> **The next commodity opportunity may emerge wherever AI-driven physical demand meets a constrained resource.**

That resource could be electricity, copper, transformers, uranium, cooling equipment, water treatment or water rights.

The research objective is to identify the bottleneck **before the market fully prices it in**.

Research Agenda: What to Track Next

The next edition of this research should build six primary charts:

1. **Crude oil:** Brent/WTI price action alongside Russian export volumes.
2. **Commodity basket:** Copper, zinc, tin, silver and crude oil, indexed to April 2025 = 100.
3. **AI electricity:** Global and Indian data-centre electricity demand forecasts.
4. **Nuclear:** India's current nuclear capacity versus the 100 GW 2047 target.
5. **Water:** NQH2O price history, clearly labelled as California-specific.
6. **Infrastructure chain:** AI → power → grid → metals → cooling → water.

These charts would add substantially more analytical value than generic stock photographs.

Sources and References

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Source Note

The central narrative, terminology and sequence of ideas in this report are based on the user's original three-page "Dark Gold → Digital Gold" draft. The draft's first page discusses crude-oil sanctions, Russian crude flows and the price action of copper, zinc, tin and silver. Its second page develops the zinc, tin, SMR and water arguments, while its third page raises the question of how water could be traded through CME. These elements have been retained as the conceptual foundation of the report.

Several claims have deliberately been presented as hypotheses rather than facts where the original draft did not establish a sufficiently strong causal link.

This document is a research note, not investment advice.